

5-Mark Questions & Answers — DBMS Module 2: Relational Algebra & Calculus

Q1. Explain all fundamental operations of Relational Algebra with notation and examples.

1. **Selection (σ):** $\sigma_{\text{Dept_ID} = 10}(\text{Employee})$ — Selects rows where Dept_ID is 10.
2. **Projection (π):** $\pi_{\{\text{Name}, \text{Salary}\}}(\text{Employee})$ — Projects Name and Salary columns.
3. **Union ($\cup$):** $R \cup S$ — Combines tuples from R and S without duplicates.
4. **Set Difference ($-$):** $R - S$ — Tuples in R but not in S.
5. **Cartesian Product ($\times$):** $R \times S$ — Pairwise combination of all rows.
6. **Rename (ρ):** $\rho_{\text{NewName}}(R)$ — Renames relation or attributes.

Q2. Explain Outer Join operations (Left, Right, Full) with concrete table diagrams.

- **Left Outer Join ($\bowtie$):** Retains all tuples from the left table; unmatched right attributes are filled with **NULL**.
- **Right Outer Join ($\bowtie$):** Retains all tuples from the right table; unmatched left attributes are filled with **NULL**.
- **Full Outer Join ($\bowtie$):** Retains all tuples from both tables, filling unmatched columns on either side with **NULL**.

Q3. Explain Domain Relational Calculus (DRC) and compare TRC vs DRC.

- **DRC Form:** $\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$ where variables represent attribute domains rather than full tuples.
- **Comparison:** TRC binds variables over entire tuples (t), while DRC binds variables over individual attribute values (x, y, z).